Propagation of mean and covariance through non-linearity y = f(x)

2D example: conversion: polar to rectangular coordinates

2D random variable $x = [\, r \;\; \theta \,]^T$

$r \sim$ Uniform[$1-\Delta r \;\; 1+\Delta r$], $\Delta r = 0.01$

$\theta \sim$ Uniform[$\pi/2-\Delta\theta \;\; \pi/2+\Delta\theta$], $\Delta\theta = 0.35$

$y = f(x)$ →

2D random variable

$y = [\, y_1 \; y_2 \,]^T$

$= [\, r\cos(\theta) \;\; r\sin(\theta) \,]^T$

EKF approach: linearize y = f(x) to estimate mean and covariance of y !

Explanation of ellipses: (graphical representation of covariance matrix)

Covariance matrix P_y (symmetric and positive semi-definite)

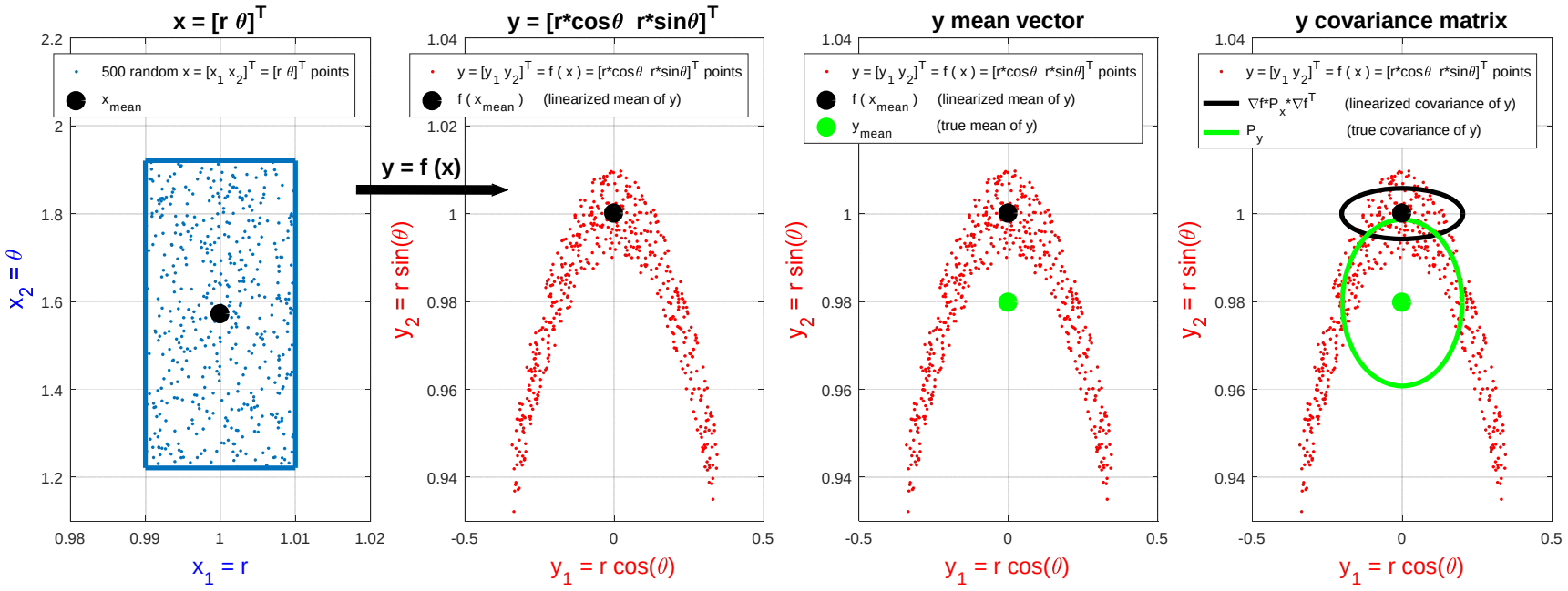
represents a quadratic form $(y-y_{mean})^T P_y^{-1} (y-y_{mean}) = 1$

This is the equation of an ellipsis in the y-plane with

center = y_{mean}

axes half-lengths = $\sqrt{\lambda_i}$, eigenvalues of P_y

axes directions = e_i, eigenvectors of P_y



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UT: propagate sigma-points thru f, estimate mean and covariance of y !

Explanation of ellipses: (graphical representation of covariance matrix)

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